Def. Let X_i $(i \in I)$ be Topological spaces with Topology $\mathcal{T}_i$, respectively. Define a Product subbasis on $\prod_{i \in I} X_i$ as;

$$S_p := \{ p_i^{-1}[U_i] \mid i \in I,\ U_i \in \mathcal{T}_i \}$$

The Topology $\mathcal{T}_p$ generated by S_p is called a Product Topology on $\prod_{i \in I} X_i$

Clearly, the product Topology is the smallest Topology s.t all Projections are Continuous. Moreover, on the Product space, all Projections are Continuous Open onto map.

Thm. Let X, Y_i $(i \in I)$ be Topological Spaces. Define a map

$$f : X \to \prod_{i \in I} Y_i \ ;\ x \mapsto (f_i(x))_{i \in I} \quad \text{where } f_i : X \to Y_i$$

Then, f is Continuous if and only if for all $i \in I$, f_i is Continuous.

Proof If f is continuous, then $f_i = p_i \circ f$ is Continuous, being Composition of Continuous. Conversely, let $B \in \mathcal{B}_p$ be an base element of $\mathcal{T}_p$. Then, by definition,

$$B = \bigcap_{i=1}^{n} p_i^{-1}[U_i] \quad \text{for some } U_i \in \mathcal{T}_i,\ i = 1, \cdots, n.$$

Now,

$$f^{-1}[B] = f^{-1}\left[\bigcap_{i=1}^{n} p_i^{-1}[U_i]\right] = \bigcap_{i=1}^{n} (f^{-1} \circ p_i^{-1})[U_i] = \bigcap_{i=1}^{n} (p_i \circ f)^{-1}[U_i] = \bigcap_{i=1}^{n} f_i[U_i]$$

$\blacksquare$

「Projection

「Embbeding

「Lemma.

Let X, Y be Topological Spaces, and let $y_0 \in Y$ be fixed. Define an Embbeding

$$E_x : X \to X \times Y : x \mapsto (x, y_0)$$

Then the E_x is an Embedding.

Proof. Let $\mathcal{U} \subseteq X$ and $V \subseteq Y$ be open sets. Then, $\mathcal{U} \times V \subseteq X \times Y$ is open set of the Product space. Then, the preimage of E_{y_0} is:

$$E_x^{-1}[\mathcal{U} \times V] = \begin{cases} \mathcal{U} & \text{if } y_0 \in V \\ \emptyset & \text{if } y_0 \notin V \end{cases}$$

Either case, the preimage is open set in X.

And, given an open set $\mathcal{U} \subseteq X$, the image

$$E_x[\mathcal{U}] = \mathcal{U} \times \{y_0\} = (\mathcal{U} \times Y) \cap (X \times \{y_0\})$$

is open set of $X \times \{y_0\}$. Finally, the injectivity is clear, So proved. ☒

「Thm.

Let $f : X \times Y \to Z$ be a Continuous map. Then, for fixed $y_0 \in Y$, the map

$$f_x : X \to Z : x \mapsto f(x, y_0)$$

is Continuous.

Proof. Since $f_x = f \circ E_x$ where E_x is the embbeding for fixed $y_0 \in Y$, it is Continuous, being the Composition of Continuous maps. ☒

「Thm. Let X_i $(i \in I)$ be Topological spaces, $A_i \subset X_i$. Then,

1) $\overline{\prod_{i \in I} A_i} = \prod_{i \in I} \overline{A_i}$

2) $\left(\prod_{i \in I} A_i\right)^\circ \subseteq \prod_{i \in I} A_i^\circ$, the equality holds when the product is finite.

Proof. $(x_i) \in \overline{\prod_{i \in I} A_i} \iff \forall \mathcal{U} \in \beta_p$ with $(x_i) \in \mathcal{U}$, $\mathcal{U} \cap \prod_{i \in I} A_i \neq \phi$

$\iff \forall \mathcal{U} \in \beta_p$ with $(x_i) \in \mathcal{U}$, $\prod_{i \in I} \pi_i[\mathcal{U}] \cap \prod_{i \in I} A_i = \prod_{i \in I} \pi_i[\mathcal{U}] \cap A_i \neq \phi$

$\iff \forall i \in I$, $\forall \mathcal{U}_i \in \beta_i$ with $x_i \in \mathcal{U}_i$, $\mathcal{U}_i \cap A_i \neq \phi$

$\iff (x_i) \in \prod_{i \in I} \overline{A_i}$

(I finite)

$(x_i) \in \left(\prod_{i \in I} A_i\right)^\circ \overset{(*)}{\iff} \exists \mathcal{U} \in \beta_p$ s.t. $(x_i) \in \mathcal{U} \subseteq \prod_{i \in I} A_i$

$\iff \forall i \in I$, $x_i \in P_i[\mathcal{U}] \subseteq A_i$

$\iff \forall i \in I$, $x_i \in A_i^\circ$.

But, if I is Not finite, $(*)$ is broke.